

# SDA GIS Full Manual Review Methodology (May 2025)

*Written by Payman Alemi*

## Table of Contents

|       |                                                                                                                      |    |
|-------|----------------------------------------------------------------------------------------------------------------------|----|
| 1     | ArcGIS Pro Portion.....                                                                                              | 1  |
| 1.1   | Initial Setup .....                                                                                                  | 1  |
| 1.2   | Convert the eWRIMS flat file to a shapefile.....                                                                     | 5  |
| 1.3   | Add the PLSS information from the BLM PLSS shapefile to the eWRIMS PODs .....                                        | 7  |
| 1.4   | Filter to just the PODs in your watershed.....                                                                       | 10 |
| 1.4.1 | Create a shapefile that just intersects the watershed .....                                                          | 10 |
| 1.4.2 | Relate the RR_Section_Intersect_BLM_2024 shapefile to the<br>ewrims_statewide_pods_PLSS_spatial_join shapefile ..... | 13 |
| 1.5   | Determine Edge Cases for Manual Review .....                                                                         | 17 |
| 1.5.1 | Create the watershed boundary layer .....                                                                            | 17 |
| 1.5.2 | Create the watershed interior buffer, interior, and exterior buffer layers.....                                      | 18 |
| 1.5.3 | Eliminating Unnecessary PODs from the Manual Review .....                                                            | 20 |
| 2     | Manual Review Process using ArcGIS and Excel .....                                                                   | 22 |
| 2.1   | Import the Manual Review POD List into Excel .....                                                                   | 22 |
| 2.2   | Bulk Download all eWRIMS documents for your watershed .....                                                          | 23 |
| 2.3   | How to fill out the manual review spreadsheet .....                                                                  | 23 |
| 3     | Appendix .....                                                                                                       | 28 |
| 3.1   | Using Power Query and M Scripts .....                                                                                | 28 |
| 3.2   | Using USGS StreamStats Tool .....                                                                                    | 31 |
| 3.3   | Using EarthPoint to convert Northings/Easting into Latitude/Longitude.....                                           | 34 |
| 3.4   | Finding POD Coordinates in the absence of latitude/longitude and northing/easting .....                              | 36 |

## 1 ArcGIS Pro Portion

### 1.1 Initial Setup

- 1) Create a local geodatabase just for this project, so that you have local storage space for you GIS files—this will substantially expedite processing in ArcGIS Pro. Normally, if you create a project in ArcGIS Pro, then ArcGIS Pro will automatically create an associated local geodatabase, which you can see in the *Contents* pane in Figure 1.

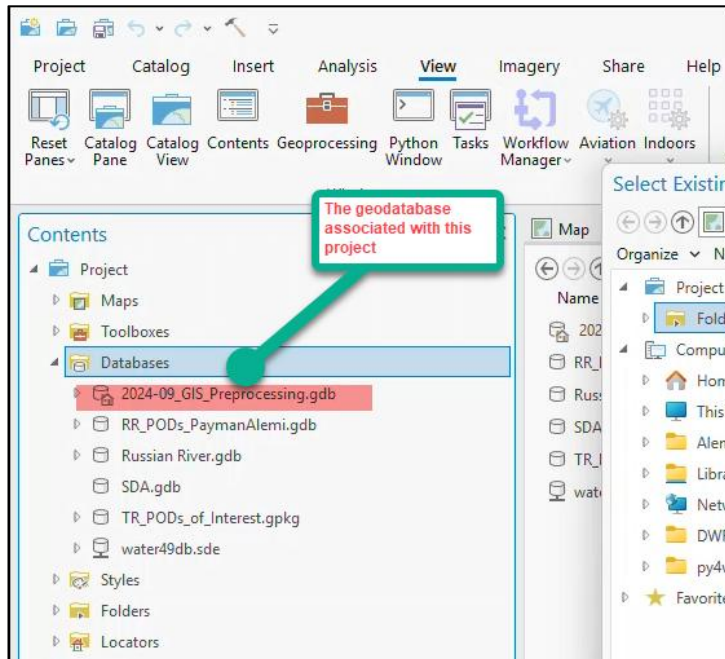


Figure 1. The geodatabase tied to the current project map.

- 2) Add the Flat\_File\_eWRIMS-YYYY-MM-DD.csv (“eWRIMS flat file”) generated by the GIS\_POD\_Flat\_File\_Prep.R script to your project map and to your local geodatabase.
- 3) Using File Explorer, navigate to the SDA\_Public\_gdb.zip file in the DWRAT\_DataScraping/Demand/InputData folder. Right-click to unzip the file and save the entire geodatabase to your computer as SDA\_public.gdb—keep it off SharePoint and/or other cloud services to avoid syncing issues.
- 4) In ArcGIS Pro, navigate to the *View* pane and select *Catalog Pane* and scroll to *Databases*:

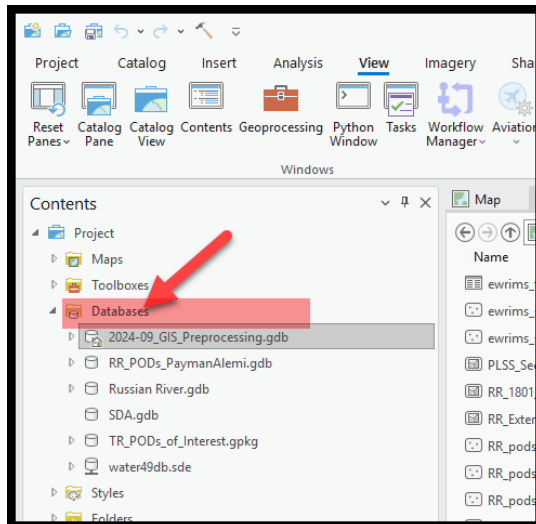


Figure 2. Database button in Catalog Pane.

- 5) Right-click on *Databases* and select *Add Database* from the pop-up menu:

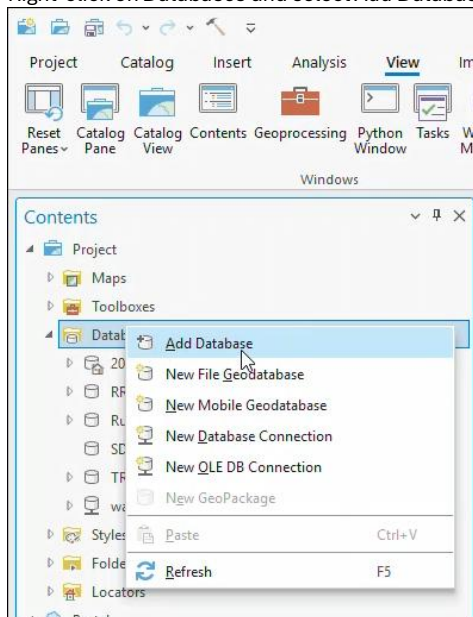


Figure 3. Adding an existing database in ArcGIS Pro.

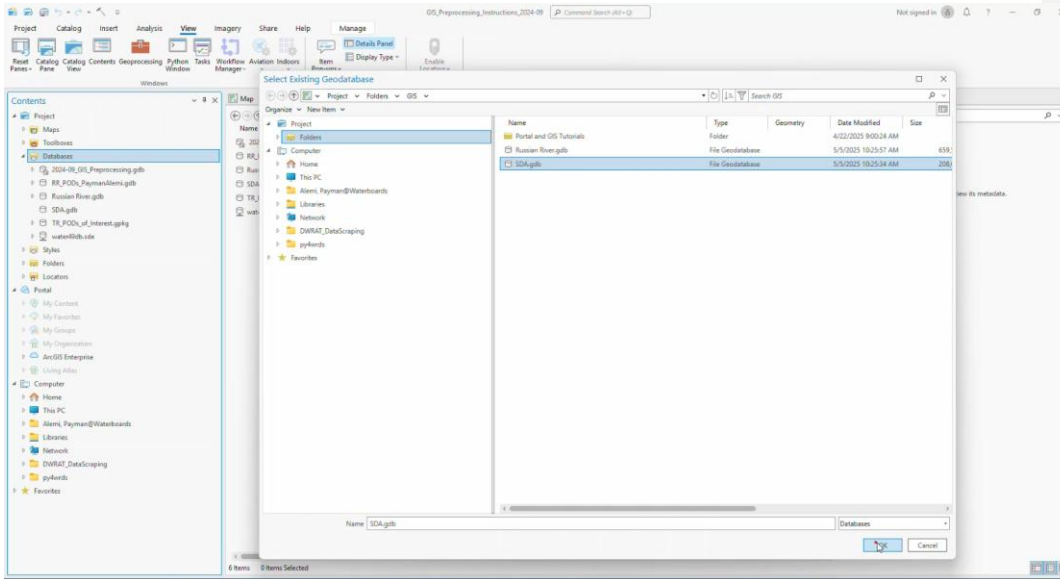


Figure 4. Select SDA\_public.gdb from you saved location.

- 6) Then navigate to *SDA\_public.gdb* in the pop-up window, click on it, and click **OK** to load it into your list of databases:
- 7) In the ArcGIS Pro ribbon, go to **View** → **Catalog** view and then navigate to the **Contents** pane. You should see **Databases**—right-click on it to reveal *SDA\_public.gdb*.
- 8) Click on *SDA\_public.gdb* and ArcGIS Pro will reveal all the shapefiles contained in the geodatabase like so:

**Commented [PA1]:** @Aleml, Payman@Waterboards  
Replace the screenshot when you create SDA\_public.gdb

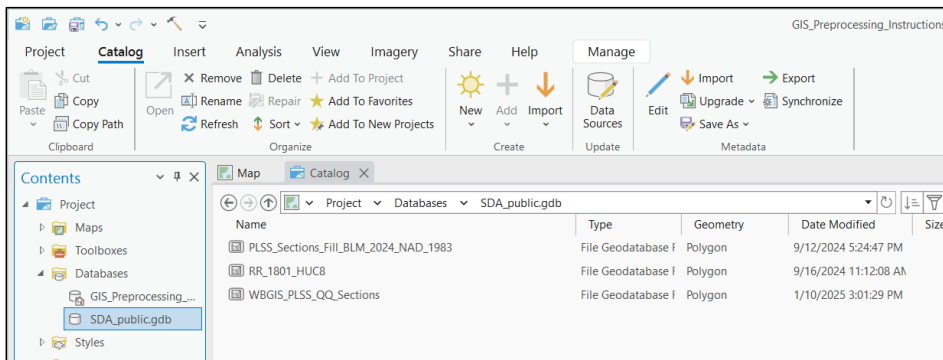


Figure 5. Shapefiles within SDA\_public.gdb

- 9) Right-click on the BLM PLSS Quarter-Quarter sections shapefile (*WBGIS.PLSS\_QQ\_Sections*) from *SDA\_public.gdb* and select **Add to Map** from the pop-up menu to add this shapefile to your project map and your local geodatabase:

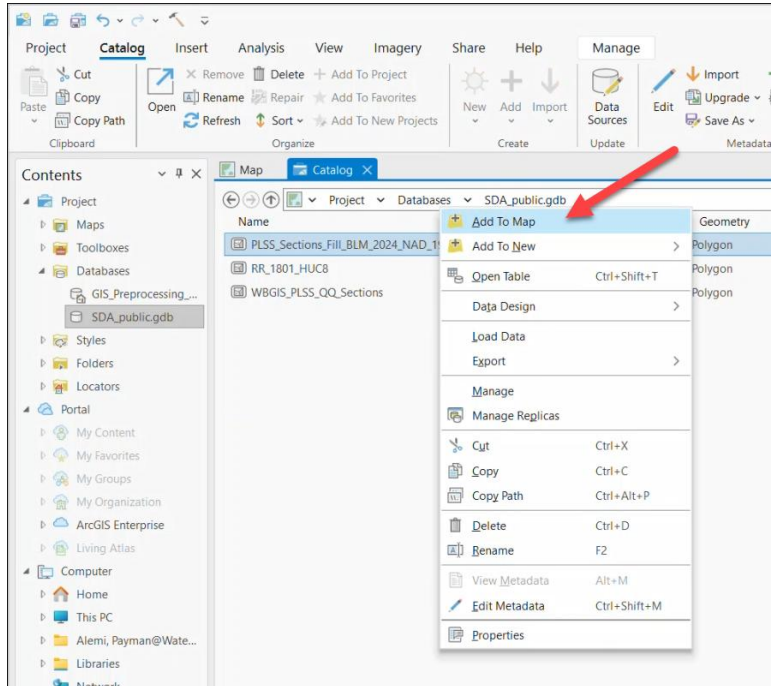


Figure 6. Adding a shapefile from a geodatabase.

- 10) Add the `PLSS_Sections_Fill_BLM_2024_NAD_1983` shapefile to your project map too—this shapefile contains the public land survey sections, townships, and ranges for the state of California.
- 11) Add the shapefile corresponding to your watershed to the project map and local geodatabase. If you would like to use the Russian River watershed as an example, add the `RR_1801_HUC8` shapefile from `SDA_public.gdb` to your project map.

## 1.2 Convert the eWRIMS flat file to a shapefile

- 1) Run the *Export Table* Geoprocessing tool. This tool allows you to change the data types for the fields in the eWRIMS flat file. At the outset, the default data type for all fields is text, which will prevent us from applying the later steps.
  - a) Find the tool by navigating to the **Analysis Pane**, clicking on the **Tools** icon, and then searching for *Export Table*. Select *Export Table (Conversion Tools)* from the resulting menu. I recommend adding the tool as a project favorite (right click and select “Add to my Favorites”), so you don’t have to search for it every time:

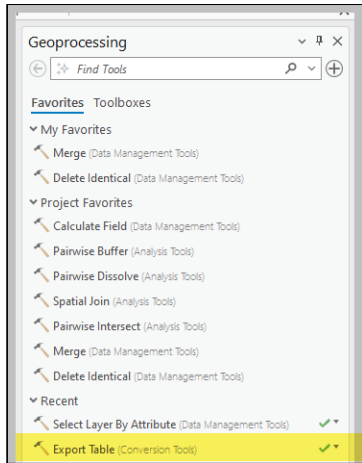


Figure 7. Export Table Tool

- Enter these parameters into the *Export Table* tool:  
**Input Table:** Flat\_File\_eWRIMS\_YYYY-MM-DD.csv  
**Output Table:** ewrims\_flat\_file\_table (save this shapefile to your local geodatabase).
- Click on the *Fields* button to expand the *Field Map*. Then click on individual fields as shown below to change the *type*.

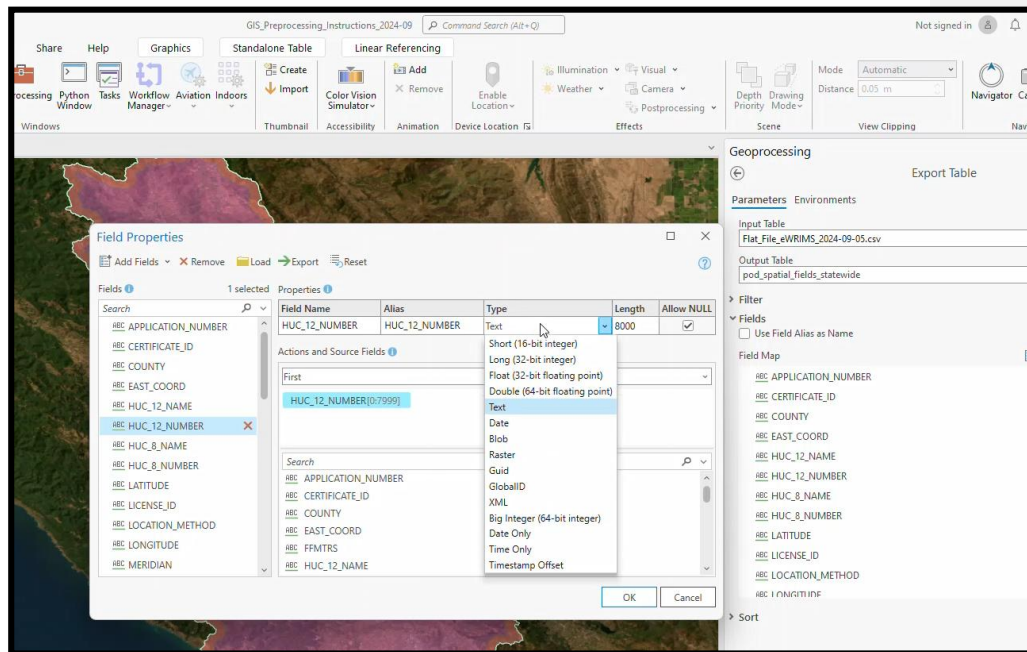


Figure 8. Adding Fields to an Attribute Table

## 7 of 38

- d) Change these fields as follows:
- HUC12\_NUMBER:** change type to *Long (32-bit integer)*
  - HUC8\_NUMBER:** change type to *Long (32-bit integer)*
  - LATITUDE:** change type to *Double (64-bit floating point)*
  - LONGITUDE:** change type to *Double (64-bit floating point)*
  - RANGE\_NUMBER:** change type to *Short (16-bit integer)*
  - SECTION\_NUMBER:** change type to *Short (16-bit integer)*
  - TOWNSHIP\_NUMBER:** change type to *Short (16-bit integer)*
- e) Click *Run* to generate the shapefile
- 2) Run the *XY Table to Point* geoprocessing tool—this tool actually converts *ewrims\_flat\_file\_table* into a shapefile.
- a) In the ribbon, navigate to *Analysis* → *Tools* → *XY Table to Point (Data Management Tools)*
  - b) Enter the following parameters and click *Run*. Figure 9 shows a screenshot for clarity, but the written instructions take precedence because it was impractical to retake a screenshot every time the instructions were slightly tweaked.

**Input Table:** ewrims\_flat\_file\_table

**Output Feature Class:** ewrims\_statewide\_pods

**X Field:** LONGITUDE

**Y Field:** LATITUDE

**Z Field:** Leave blank

**Coordinate System:** **GCS\_North\_American\_1983**—which you can click on the globe and navigate to select; it may appear with the alias *NAD 1983*.

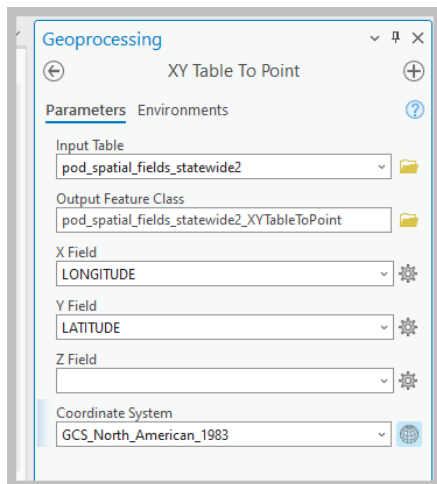


Figure 9. XY Table to Point Tool

### 1.3 Add the PLSS information from the BLM PLSS shapefile to the eWRIMS PODs

- 1) Spatially join the PLSS BLM shapefile to the ewrims\_statewide\_pod shapefile

## 7 of 38

## 8 of 38

- In the ribbon, navigate to *Analysis* → *Tools* → *Spatial Join (Analysis Tools)*
- Enter these parameters and click *Run*. The screenshot shows the tool's appearance, but the written instructions below take precedent:

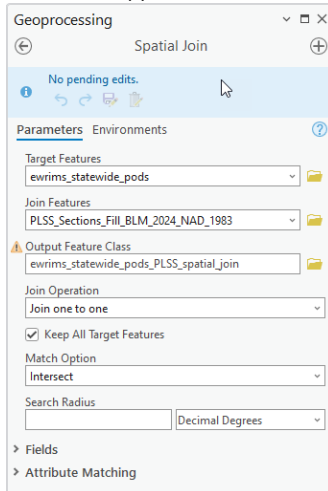


Figure 10. Spatial Join Tool

**Target Features:** ewrims\_statewide\_pods

**Join Features:** PLSS\_Sections\_Fill\_BLM\_2024\_NAD\_1983

**Output Feature Class:** ewrims\_statewide\_pods\_PLSS\_spatial\_join

**Join Operation:** One to Many

**Checkmark:** Keep All Target Features

**Match Option:** Intersect

**Search Radius:** Leave the quantity field blank but select **Decimal Degrees** for the units.

- Right-click on the *ewrims\_statewide\_pods\_PLSS\_spatial\_join* shapefile and open the attribute table. Check that both the *MTRS* and *FFMTRS* fields are present.
- Add the *MTRS\_Match* field to the attribute table by clicking the *Add Field* button.

| OBJECTID | Shape | Join_Count | TARGET_FID | APPLICATION_NUMBER | CERTIFICATE_ID | COUNTY      | EAST_COORD     | HUC_12_NAME              | HUC_12_NUMBER | HUC_8_NAME           |
|----------|-------|------------|------------|--------------------|----------------|-------------|----------------|--------------------------|---------------|----------------------|
| 1        | Point | 1          | 1          | A000016            | 000041         | Los Angeles | 6474612.147768 | Headwaters Amargosa...   | 18090206      | Antelope-Fremont Vi  |
| 2        | Point | 1          | 2          | A000016            | 000041         | Los Angeles | 6474612.147768 | Headwaters Amargosa...   | 18090206      | Antelope-Fremont Vi  |
| 3        | Point | 1          | 3          | A000016            | 000041         | Los Angeles | 6474614.806227 | Headwaters Amargosa...   | 18090206      | Antelope-Fremont Vi  |
| 4        | Point | 1          | 4          | A000016            | 000041         | Los Angeles | 6474612.147768 | Headwaters Amargosa...   | 18090206      | Antelope-Fremont Vi  |
| 5        | Point | 1          | 5          | A000018            | 002871         | Glenn       | 6547331        | Murphy Slough-Sacra...   | 18020157      | Big Chico Creek-Sacr |
| 6        | Point | 0          | 6          | NA                 | NA             | NA          | 0              | NA                       | 0             | NA                   |
| 7        | Point | 0          | 7          | NA                 | NA             | NA          | 0              | NA                       | 0             | NA                   |
| 8        | Point | 1          | 8          | A000023            | 001986         | Madera      | 6793175        | Lost Lake-San Joaquin... | 18040001      | Middle San Joaquin   |
| 9        | Point | 1          | 9          | A000026            | 000036         | Lake        | 6346157.876052 | Kelsey Creek             | 18020116      | Upper Cache          |
| 10       | Point | 1          | 10         | A000027            | 003165         | Colusa      | 6574809.313105 | Packer Lake-Sacrament... | 18020104      | Sacramento-Stone C   |
| 11       | Point | 1          | 11         | A000027            | 003165         | Colusa      | 6564109.470004 | Packer Lake-Sacrament... | 18020104      | Sacramento-Stone C   |
| 12       | Point | 1          | 12         | A000027            | 003165         | Glenn       | 6559684.850572 | Packer Lake-Sacrament... | 18020104      | Sacramento-Stone C   |
| 13       | Point | 1          | 13         | A000042            | 000211         | Mono        | 7092376.316981 | Lower Rock Creek         | 18090102      | Crowley Lake         |

Figure 11. Add MTRS\_Match Field

## 8 of 38

- 4) Then enter these parameters for the *Field Name*, *Alias*, and *Data Type* and save the changes to the attribute table.

| Visible                             | Read Only                | Field Name           | Alias              | Data Type | Allow NULL                          | Highlight                | Number Format | Domain | Default | Length |
|-------------------------------------|--------------------------|----------------------|--------------------|-----------|-------------------------------------|--------------------------|---------------|--------|---------|--------|
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | SHAPE_LENGTH         | SHAPE_LENGTH       | Double    | <input checked="" type="checkbox"/> | <input type="checkbox"/> | Numeric       |        |         |        |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | TOWNSHIP_NORM        | TOWNSHIP_NORM      | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 5      |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | RANGE_NORM           | RANGE_NORM         | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 5      |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | TOWNSHIP             | TOWNSHIP           | Double    | <input checked="" type="checkbox"/> | <input type="checkbox"/> | Numeric       |        |         |        |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | TOWNSHIP_DIR         | TOWNSHIP_DIR       | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 1      |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | RANGE                | RANGE              | Double    | <input checked="" type="checkbox"/> | <input type="checkbox"/> | Numeric       |        |         |        |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | RANGE_DIR            | RANGE_DIR          | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 1      |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | SECTION_             | SECTION_           | Short     | <input checked="" type="checkbox"/> | <input type="checkbox"/> | Numeric       |        |         |        |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | SECTION_CLASSIFIER_1 | SECTION_CLASSIFIER | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 50     |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | MERIDIAN_CD          | MERIDIAN_CD        | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 10     |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | MTRS                 | strSection         | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 255    |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> | MTRS_Match           | MTRS_Match         | Text      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |               |        |         | 255    |

Figure 12. Field Properties of MTRS\_Match

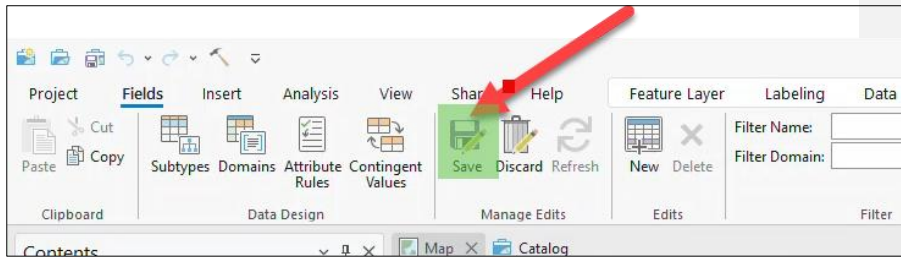


Figure 13. Saving changes to an attribute table

- 5) Back in the attribute table, right-click on the *MTRS\_Match* field, and click on *Calculate Field*:

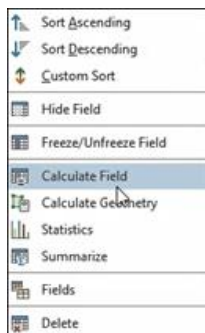


Figure 14. Calculate Field for MTRS\_Match

- 6) Calculate **MTRS\_Match** by using the Python expression below and set the data type to text; otherwise, the field calculator will fail.

## 10 of 38

Select *Python* as the **Expression Type**. Toggle on the *Enable Undo* button in case the code fails and click *OK*.

```
MTRS_Match = reclass(!FFMTRS!,!MTRS!)
```

**Code Block**

```
def reclass(FFMTRS, MTRS):  
    if FFMTRS == MTRS:  
        return "Y"  
    else:  
        return "N"
```

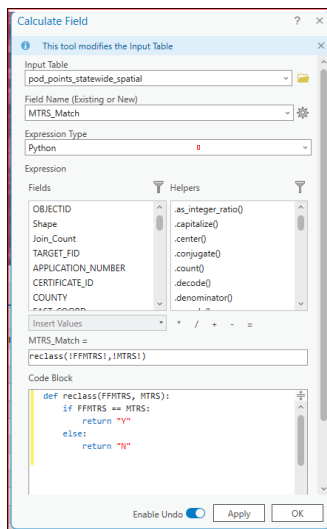


Figure 15. Python code block for MTRS\_Match

### 1.4 Filter to just the PODs in your watershed

You can apply the steps in this section to any watershed. Just use the shapefiles that correspond to your watershed instead, but the sequence is the same. The example here uses the Russian River watershed.

#### 1.4.1 Create a shapefile that just intersects the watershed

- 1) Select *PLSS\_Section\_Fill\_BLM\_2024\_NAD\_1983* by location. In the ribbon, navigate to *Map* → *Select by Location*
- 2) Enter these parameters into the *Select by Location* wizard and click *OK*.  
*Input Features:* PLSS\_Sections\_Fill\_BLM\_2024\_NAD\_1983  
*Relationship:* Intersect  
*Selecting Features:* RR\_1801\_HUC8 (or the polygon shapefile that represents your watershed)  
*Search Distance:* Leave the quantity blank and enter **Decimal Degrees** for the units  
*Selection:* New Selection

## 10 of 38

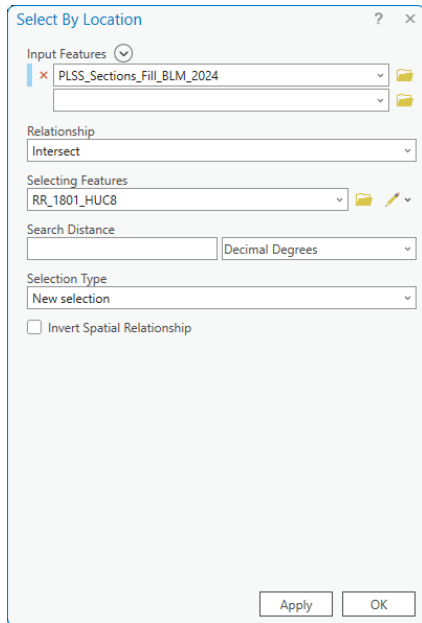


Figure 16. Select by Location to your watershed

- 3) The portion of the *PLSS\_Sections\_Fill\_BLM\_NAD\_1983* shapefile that overlaps with the *RR\_1801\_HUC8* shapefile should be selected. Right-click on the *PLSS\_Sections\_Fill\_BLM\_NAD\_1983* shapefile in the *Contents* pane and click on *Selection* → *Make Layer from Selected Features*.
- 4) Rename the newly created layer *XX\_Section\_Intersect\_BLM\_2024* with XX representing your watershed alias. For the Russian River, we are using RR—hence, we are renaming the file *RR\_Section\_Intersect\_BLM\_2024*.
- 5) Save *XX\_Section\_Intersect\_BLM\_2024* to your local geodatabase. Right-click on the shapefile and go to *Data* → *Export Features* and a dialog box will appear.
- 6) Input these parameters into the *Export Features* dialog box
  - a. *Input Features*: *XX\_Section\_Intersect\_BLM\_2024*
  - b. *Output Feature Class*: *XX\_Section\_Intersect\_BLM\_2024*

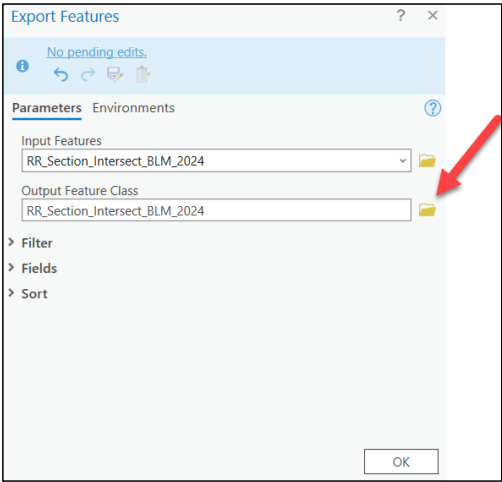


Figure 17. Export Features Dialog Box

The output feature class should be saved by default to your local geodatabase but to verify, you can click on the folder icon next to the field, navigate to the geodatabase, and select it from the search window that emerges:

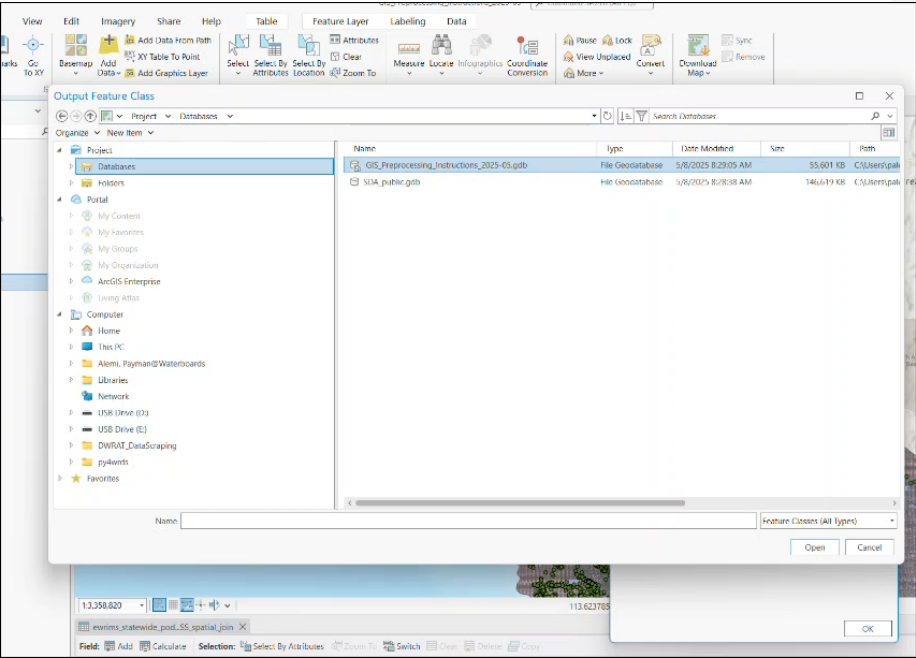
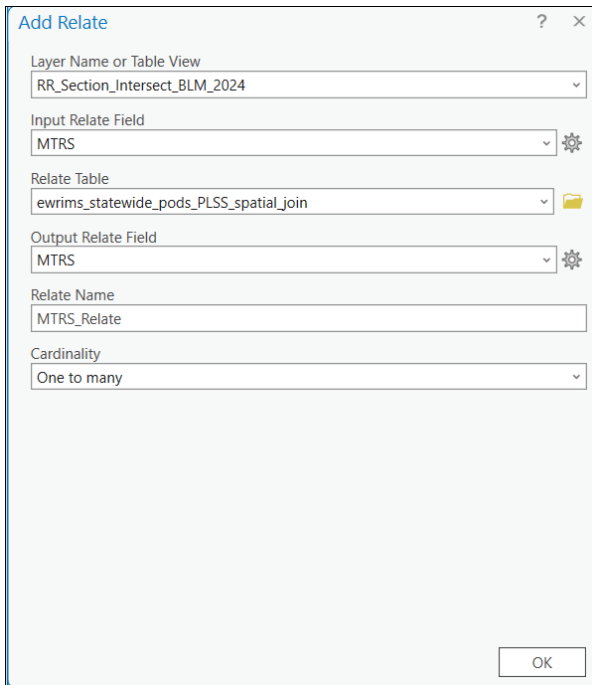


Figure 18. Export Features Search Window

## 13 of 38

### 1.4.2 Relate the XX\_Section\_Intersect\_BLM\_2024 shapefile to the ewrims\_statewide\_pods\_PLSS\_spatial\_join shapefile

- 1) Right-click **XX\_Section\_Intersect\_BLM\_2024** and navigate to *Joins and Relates* → *Add Relate*
- 2) Input these parameters into the *Add Relate* wizard and click *OK*:  
*Layer Name: or Table View:* RR\_Section\_Intersect\_BLM\_2024  
*Input Relate Field:* MTRS  
*Relate Table:* ewrims\_statewide\_pods\_PLSS\_spatial\_join  
*Output Relate Field:* MTRS  
*Relate Name:* MTRS\_Relate  
*Cardinality:* One to many



The screenshot shows the 'Add Relate' dialog box with the following settings:

- Layer Name or Table View:** RR\_Section\_Intersect\_BLM\_2024
- Input Relate Field:** MTRS
- Relate Table:** ewrims\_statewide\_pods\_PLSS\_spatial\_join
- Output Relate Field:** MTRS
- Relate Name:** MTRS\_Relate
- Cardinality:** One to many

The 'OK' button is located at the bottom right of the dialog.

Figure 19. Adding MTRS\_Relate

- 3) Right-click on the *RR\_Section\_Intersect\_BLM\_2024* layer in the *Contents* pane and go to *Properties* → *Selection*
  - A. Check the box to *automatically select related data*
  - B. Click *OK*.

## 13 of 38

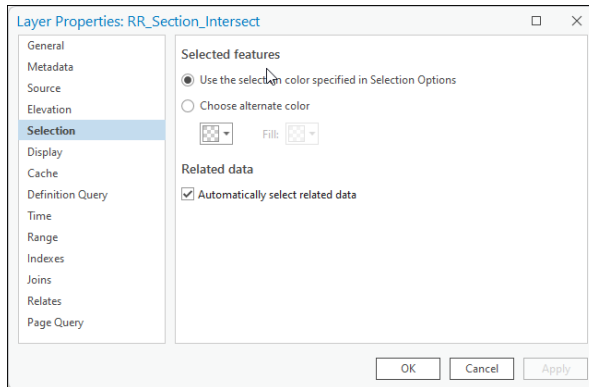
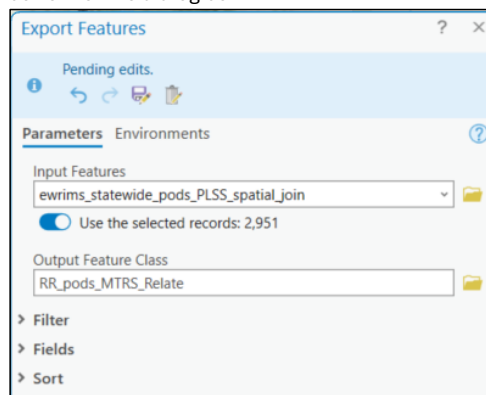


Figure 20. Automatically selecting related data

- 4) **Create *XX\_pods\_MTRS\_Relate* shapefile** (remember XX is your watershed alias; for the Russian River, it's RR)
  - A. Right-click the *XX\_Section\_Intersect\_BLM\_2024* layer in the *Contents* pane and click on the *Attribute Table*.
  - B. Select all the polygons in the *XX\_Section\_Intersect\_BLM\_2024* layer by clicking anywhere on the attribute table and clicking **Ctrl+A** on your keyboard.
  - C. Since we checked the *Automatically select related data* checkbox earlier, all the related PODs in the *ewrims\_statewide\_pods\_PLSS\_spatial\_join* layer will also be selected.
  - D. Right-click the *ewrims\_statewide\_pods\_PLSS\_spatial\_join* layer in the *Contents* pane and go to *Data* → *Export Features*
  - E. Toggle on the *Use the selected records* button and export the feature class as *XX\_pods\_MTRS\_Relate* to your local geodatabase. Click **OK** at the bottom of the dialog box.

Figure 21. Creating the *RR\_pods\_MTRS\_Relate* layer

- F. Clear all selections from your ArcGIS Pro project.

- 5) **Create the *XX\_pods\_FFMTS\_Relate* shapefile**

## 15 of 38

Select PODs where the *FFMTRS* field in *ewrims\_statewide\_pods\_PLSS\_spatial\_join* matches the *MTRS* field in the *XX\_Section\_Intersect\_BLM\_2024* shapefile.

- A. Right-click on the *ewrims\_statewide\_pods\_PLSS\_spatial\_join* shapefile and go to *Joins and Relates* → *Remove All Relates*.
- B. Right-click on the *XX\_Section\_Intersect\_BLM\_2024* shapefile and remove all the relates.
- C. Right-click on the *XX\_Section\_Intersect\_BLM\_2024* shapefile and go to *Properties* → *Selection*. Underneath the *Related Data* section, ensure that the *Automatically select related data* checkbox is still checked.
- D. Right-click on the *XX\_Section\_Intersect\_BLM\_2024* again and go to *Joins and Relates* → *Add Relate* and enter these parameters, and then click *OK*.

*Layer Name or Table View:* *XX\_Section\_Intersect\_BLM\_2024*

*Input Relate Field:* *MTRS*

*Relate Table:* *ewrims\_statewide\_pods\_PLSS\_spatial\_join*

*Output Relate Field:* *FFMTRS*

*Relate Name:* *FFMTRS\_Relate*

*Cardinality:* *One to Many*

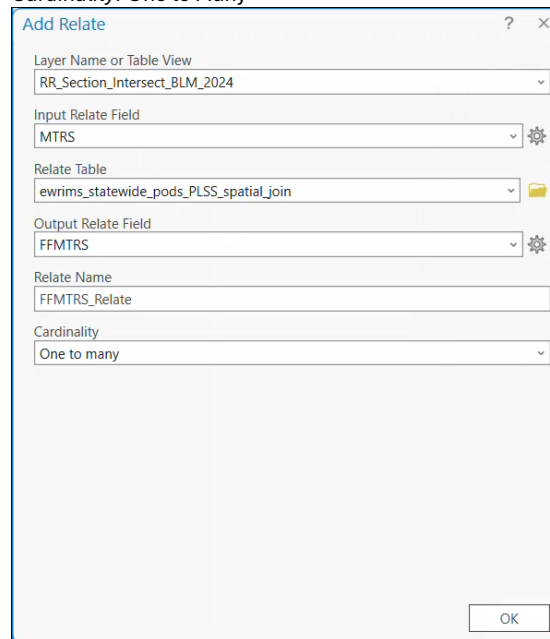


Figure 22. Creating *FFMTRS\_Relate*

- E. Open the attribute table of the *XX\_Section\_Intersect\_BLM\_2024* shapefile and select all the polygons.
- F. Right-click on the *ewrims\_statewide\_pods\_PLSS\_spatial\_join* shapefile and go to *Data* → *Export Features*
  - i. Toggle on the *use selected records* button and export as *XX\_pods\_FFMTRS\_Relate* to your local geodatabase.

## 15 of 38

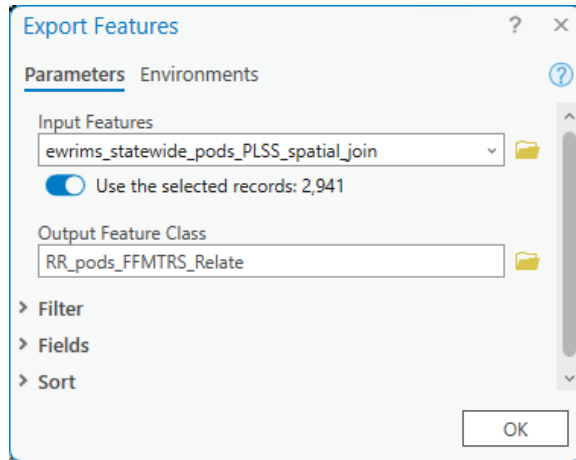


Figure 23. Creating the RR\_pods\_FFMTRS\_Relate layer

G. Clear all the selections from your ArcGIS Pro project.

- 6) Merge the XX\_pods\_MTRS\_Relate and XX\_pods\_FFMTRS\_Relate shapefiles.  
 A. Go to the *Analysis Pane* and navigate to the *Merge (Data Management Tools)* tool and enter these parameters:  
*Input datasets:* XX\_pods\_FFMTRS\_relate, XX\_pods\_MTRS\_relate  
*Output dataset:* XX\_pods\_merged

**Commented [PA2]:** @Alemi, Payman@Waterboards start here in the afternoon.

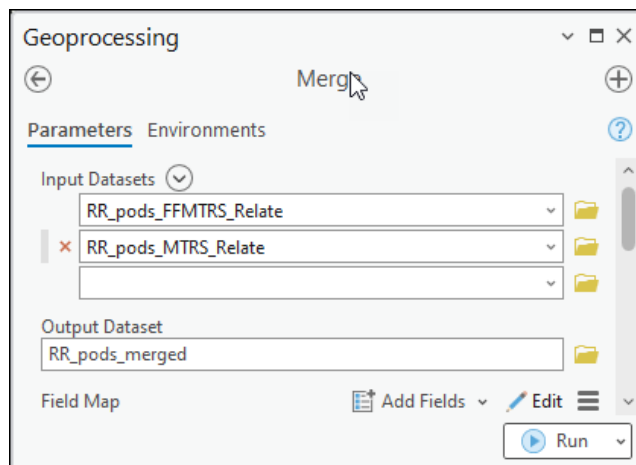


Figure 24. Merge (Data Management Tools) Tool

The total records should be the sum of the records in the attribute tables of the two layers.

- 7) Delete identical records from the XX\_pods\_merged shapefile with the *Delete Identical (Data Management Tools)* tool based on the *POD\_ID* field and click *Run*.

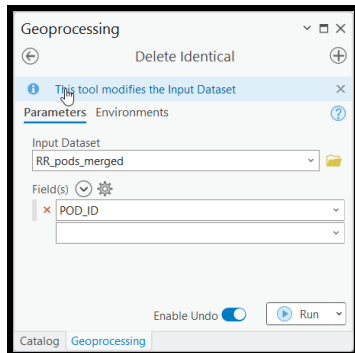


Figure 25. Delete Identical (Data Management Tools) Tool

- 8) You have now captured all PODs that geospatially may be located in your watershed. As of 5/29/2025, this series of steps yield 2,919 PODs in the *RR\_pods\_merged* shapefile. The number of PODs will vary for other watersheds; if you have thousands of PODs, a full manual review will be infeasible, so SDA recommends whittling down the list to edge cases as indicated in the next section.
- 9) Add a new text field, *Keep\_POD*, to the attribute table of the *XX\_pods\_merged* shapefile.
- 10) Export the attribute table of the *XX\_pods\_merged* shapefile as a csv and add a datestamp as the suffix, e.g., *XX\_pods\_merged\_YYYY-MM-DD.csv*, into the *DWRAT\_DataScraping/Demand/OutputData* folder. This CSV will serve as the full list of possible PODs for your watershed and after you finish your manual review, you will remove the out of watershed PODs from this CSV to create your final watershed POD List.

## 1.5 Determine Edge Cases for Manual Review

### 1.5.1 Create the watershed boundary layer

- 1) Load the *RR\_1801\_HUC8* shapefile from the SDA GIS geodatabase. If you are creating a boundary polyline for another watershed, then the corresponding HUC8 or HUC12 shapefile. The remaining steps are identical for other watersheds.
- 2) Navigate to the *Analysis Pane* → *Tools* and search for *Polygon to Line*. This tool is only licensed for use in **ArcGIS Pro – Advanced**.
- 3) Enter these parameters and click *Run*:
  - Input Features*: *RR\_1801\_HUC8* (or your watershed polygon)
  - Output Feature Class*: *XX\_WB* (save to your local geodatabase)
  - Checkmark Identify and store polygon neighboring information*.

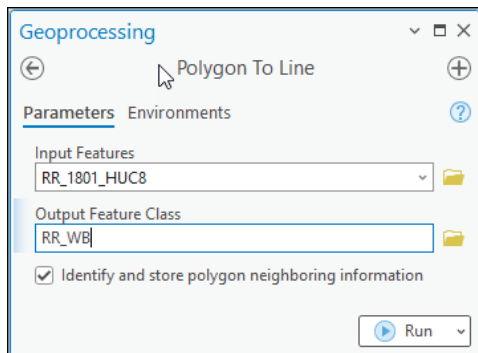


Figure 26. Polygon To Line Tool

### 1.5.2 Create the watershed interior buffer, interior, and exterior buffer layers

- 1) Navigate to the *Analysis Pane* → *Tools*, search for the *Pairwise Buffer (Analysis Tools)* tool.
- 2) Enter the following parameters and click *Run*:  
*Input Features*: XX\_WB  
*Output Feature Class*: XX\_WB\_Buffer and save to your local geodatabase.  
*Distance [value or field]*: Linear Unit, 1, Statute Miles  
*Method*: Planar  
*Dissolve Type*: No Dissolve  
*Maximum Offset Deviation*: 0, Decimal Degrees

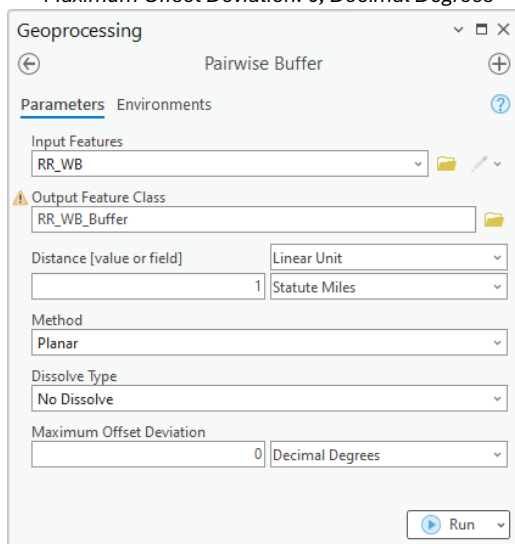


Figure 27. Pairwise Buffer (Analysis Tools) Tool

## 19 of 38

- 3) Navigate to the *Analysis Pane* → *Tools* and search for *Union*. Click on the *Union (Analysis Tools)* tool.
- 4) Enter these parameters and click *Run*:
  - Input Features*: XX\_WB\_Buffer (or the buffer layer you created for your watershed), RR\_1801\_HUC8 (or the polygon shapefile for your watershed)
  - Ranks*: Leave blank
  - Output Feature Class*: XX\_WB\_Buffer\_Union (save to your local geodatabase)
  - Attributes to Join*: All attributes
  - Gaps Allowed*: Check the box

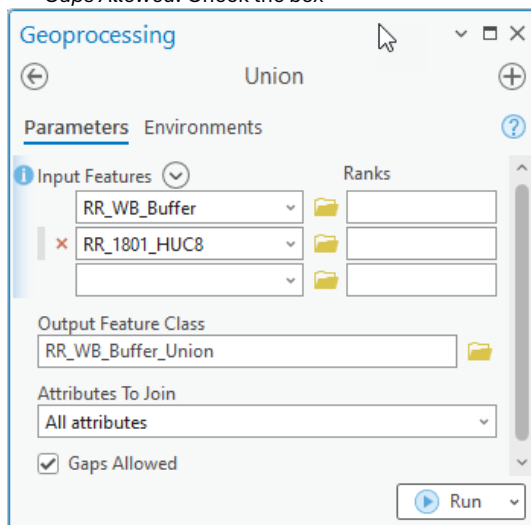


Figure 28. Union (Analysis) Tools

- 5) Right-click on the *XX\_WB\_Buffer\_Union* shapefile in the *Contents* pane and open the attribute table.
- 6) There should be 3 records in the attribute table (highlight each row to identify which is which):
  - a. **Optional**: 1 record that corresponds to the 1-miles sliver surrounding the watershed boundary. Export this record as a separate layer, *XX\_WB\_Exterior\_Buffer* to your local geodatabase.
  - b. **Mandatory**: 1 record that corresponds to the 1-mile sliver within the watershed boundary. Export this record as a separate layer, *XX\_WB\_Interior\_Buffer*, to your local geodatabase.
  - c. **Mandatory**: 1 record that corresponds to the interior polygon within the watershed, 1 mile smaller on every side than the watershed itself. Export this record as a separate shapefile, *XX\_WB\_Interior*, to your local geodatabase.

## 19 of 38

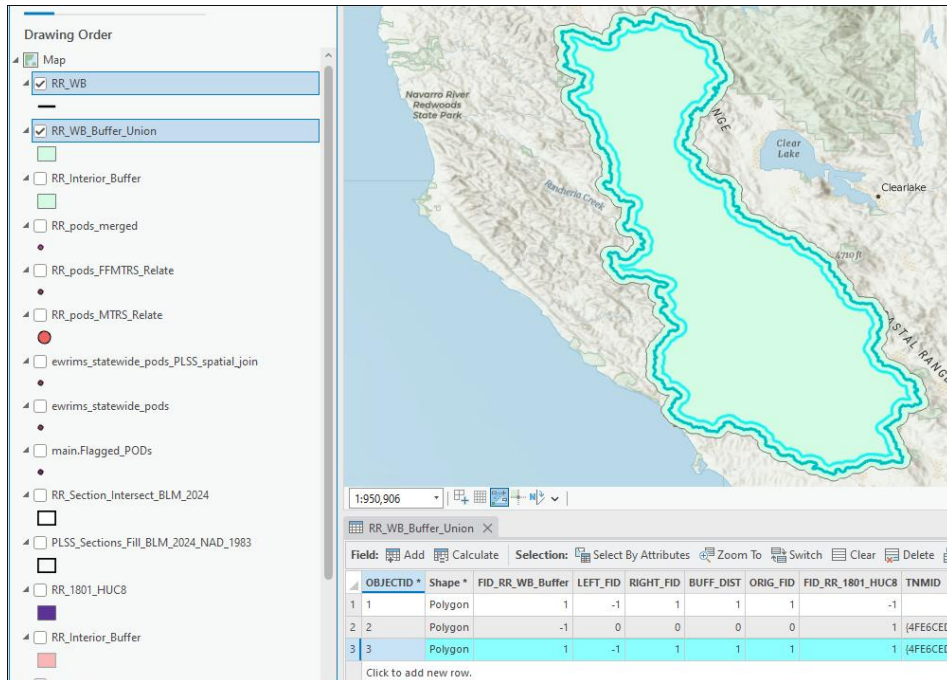


Figure 29. In this image, the “RR\_WB\_Interior\_Buffer” record is selected.

### 1.5.3 Eliminating Unnecessary PODs from the Manual Review

- 1) Right-click in the Contents pane on the *XX\_pods\_merged* shapefile and open the attribute table. Add a new field with the name and alias *Case* and data type *Short* into the attribute table.
- 2) Go to the *Analysis* tab → *Tools* → *Select Layer by Location* tool
- 3) Enter these parameters and click *Run* to execute:
  - Input Features:* *XX\_pods\_merged*
  - Relationship:* *Intersect*
  - Selecting Features:* *XX\_WB\_Interior*
  - Search Distance:* Blank and Decimal Degrees
  - Selection Type:* *New Selection*

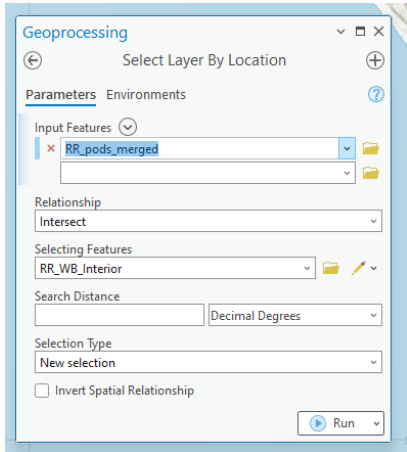


Figure 30. Select Layer by Location Tool

- 4) Once the selection is complete, open the attribute table of the *XX\_pods\_merged* shapefile and locate the *Case* field.
- 5) Use the field calculator to update the *Case* field value to 0 for the selected points
  - a. Make sure “Use the selected records” is toggled on.
- 6) Run a **definition query** to exclude all Case 0 records from *XX\_pods\_merged*
  - a. Right click layer > properties > Definition Query > Add new
  - b. Define your query such that Where *Case is null*

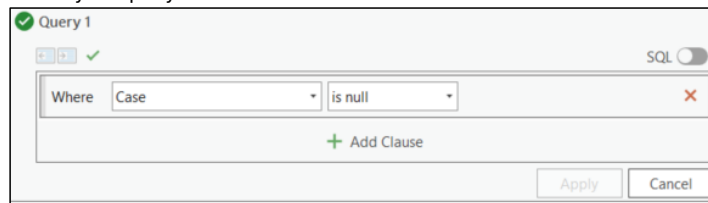


Figure 31. Definition Query for null cases

- 7) Export the selected records as a new shapefile, *XX\_pods\_manual\_review1*.
- 8) Add a definition query to *ewrims\_statewide\_pods\_PLSS\_spatial\_join* to search for any records where:
  - a. TRIB\_DESC begins with *Russian* (replace with your watershed name, “Butte”, “Napa”, etc.) OR
  - b. WATERSHED begins with *Russian* OR (replace with your watershed name)
  - c. SOURCE\_NAME begins with *Russian* (replace with your watershed name)
- 9) After activating the definition query, export the records in the *ewrims\_statewide\_pods\_PLSS\_spatial\_join* attribute table to a new shapefile, *XX\_pods\_manual\_review2*, in your local geodatabase.
- 10) Merge *XX\_pods\_manual\_review1* and *XX\_pods\_manual\_review2* into a single shapefile, *XX\_pods\_manual\_review* by using the Merge tool.
- 11) Delete the identical records from *XX\_pods\_manual\_review* based on the *POD\_ID* field.
- 12) Export *XX\_pods\_manual\_review* as a CSV, *XX\_pods\_manual\_review.csv* to the *DWRAT\_DataScraping/Demand/OutputData* folder.

## 2 Manual Review Process using ArcGIS and Excel

### 2.1 Import the Manual Review POD List into Excel

- 1) Create an `xlsx` file within your `DWRAT_DataScraping/Demand/OutputData` folder called `XX_pods_manual_review.xlsx`.
- 2) Import `XX_pods_manual_review.csv` into `XX_pods_manual_review.xlsx` (“manual review spreadsheet”) by using Power Query—refer to [Appendix 3.1 Using Power Query and M Scripts](#).
- 3) Navigate to the `RR_GIS_Manual_Review_M_Script.txt` (M Script”) file in the `Scripts` folder of your `DWRAT_DataScraping/Demand` GitHub sub-repository and open it in Notepad++.
- 4) Copy and paste all the content from the `M Script` into the Advanced Editor in Power Query.
- 5) Confirm there are no syntax errors and click *Done*.
- 6) The `M script` makes a bunch of changes to the original CSV that you should see reflected in Power Query:
  - a. LATITUDE gets renamed to `ewrims_latitude`
  - b. LONGITUDE gets renamed to `ewrims_longitude`
  - c. SOURCE\_NAME gets renamed to `ewrims_SOURCE_NAME`
  - d. TRIB\_DESC gets renamed to `ewrims_TRIB_DESC`
  - e. WATERSHED gets renamed to `ewrims_WATERSHED`
  - f. COUNTY gets renamed to `ewrims_COUNTY`

The `M Script` also adds the following fields:

- g. Report\_Latitude
  - h. Report\_Longitude
  - i. Report\_Northing
  - j. Report\_Easting
  - k. Report\_Zone
  - l. Report\_County
  - m. Report\_Source Creek
  - n. Report\_Tributary
  - o. Report\_Thence
  - p. Report\_APN
  - q. Report\_Township
  - r. Report\_Range
  - s. Report\_Section
  - t. Report\_QSection
  - u. Report\_QQSection
  - v. Report\_POD\_Information
  - w. LATITUDE
  - x. LONGITUDE
  - y. StreamStats
  - z. Keep\_POD
  - aa. POD\_Justification
- 7) After you apply the `M Script` with Power Query, export the resulting query as the “[ID]\_GIS\_Manual\_Review” sheet within your spreadsheet.

## 2.2 Bulk Download all eWRIMS documents for your watershed

- 1) Create a *Reports* child folder (if it hasn't already been created) within your *DWRAT\_DataScraping/Demand/InputData* folder. Replace [\[Watershed\]](#) with your watershed name, e.g., *Navarro*. The *eWRIMS\_Document\_Bulk\_Downloader.R* script will save all the eWRIMS documents for your watershed to this folder if you follow the remaining steps in this subsection.
- 2) Open *Demand.Rproj* with RStudio.
- 3) Navigate to your local version of the *DWRAT\_DataScraping* GitHub repository.
- 4) Open the *Watershed\_Demand\_Datasets\_Paths.xlsx* spreadsheet from the *DWRAT\_DataScraping/Demand/InputData* folder.
  - a. Scroll to the *GIS\_Preprocessing* section of the spreadsheet and fill out this field:
    - i. **EWRIMS\_REPORTS\_FOLDER\_PATH**—Enter the full file path for your *Reports* folder.
  - b. Save and close the spreadsheet.
- 5) Navigate to the *eWRIMS\_Document\_Bulk\_Downloader.R* script and run the entire script. The script is designed to skip any reports that have already been downloaded for your watershed. Consequently, you won't waste time if you have to pause the script halfway—when you resume, the script will just pick up where you left off. It is unlikely that the number of PDFs in your *Reports* folder will perfectly match the number of PODs in *RR\_pods\_merged\_2024-12-18.csv (your full POD list)* because:
  - a. Some water rights have multiple PODs and there is only 1 eWRIMS PDF document for each water right.
  - b. A few water rights only have a paper record that has not been scanned into eWRIMS yet.
  - c. A few water rights have corrupt PDFs.

## 2.3 How to fill out the manual review spreadsheet

- 1) Now that you have downloaded the eWRIMS PDFs, you can begin the manual review by referring to the PDF for each water right to fill out the various *report fields*. The figures below demonstrate where to look for each *report field*.

STATE OF CALIFORNIA  
CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY  
STATE WATER RESOURCES CONTROL BOARD

DIVISION OF WATER RIGHTS

PERMIT FOR DIVERSION AND USE OF WATER

PERMIT 20828

Application 30245 of Carolyn L. Pride

4026 Spring Mountain Road, St. Helena, CA 94574

filed on April 19, 1993, has been approved by the State Water Resources Control Board  
SUBJECT TO PRIOR RIGHTS and to the limitations and conditions of this permit.

Permittee is hereby authorized to divert and use water as follows:

1. Source:

Tributary to: Report\_Tributary

Unnamed Stream

Ritchie Creek thence

Napa River thence

San Pablo Bay Report\_Thence

Report\_Section Report\_Township Report\_Range Report\_Meridian

| 2. Location of point of diversion <u>Report_POD_Information</u>   | 40-acre subdivision of public land survey or projection thereof | Section | Township | Range | Base and Meridian |
|-------------------------------------------------------------------|-----------------------------------------------------------------|---------|----------|-------|-------------------|
| North 1,000 feet and West 1,500 feet from SE corner of Section 19 | SW $\frac{1}{4}$ of SE $\frac{1}{4}$                            | 19      | 8N       | 6W    | MD                |
| <u>Report_QQ Section</u>                                          | <u>Report_Q Section</u>                                         |         |          |       |                   |
|                                                                   |                                                                 |         |          |       |                   |
|                                                                   |                                                                 |         |          |       |                   |
|                                                                   |                                                                 |         |          |       |                   |

County of Napa

SWRCB 14 (6-94)

Report\_County

Figure 32. Locating the Report Fields on an eWRIMS document (Part 1)

Initial Statement of Water Diversion and Use  
Page 2 of 4

| Section 1: Type of Claim                                                                                                                                                                                                                              |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| (Select all that apply to the type of claim(s) under which you are diverting water)                                                                                                                                                                   |  |
| <input checked="" type="checkbox"/> Riparian <input type="checkbox"/> Pre-1914 <input type="checkbox"/> Court Decree <input type="checkbox"/> Pending Appropriative Application <input type="checkbox"/> Pueblo <input type="checkbox"/> Other: _____ |  |
| *If you checked Court Decree, Pending Appropriative Application, or Other, list the Decree Number, Application ID Number or Status or provide an explanation                                                                                          |  |
| List any related existing water rights, if applicable (e.g. Appropriative Water Right ID: A012345)                                                                                                                                                    |  |

| Section 2: Water Course Description                                                           |                                                                          |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Report_Source_Creek<br>Water Course Name at the Point of Diversion (POD)<br><b>Mill Creek</b> | Report_Tributary<br>Water Course is tributary to<br><b>Russian River</b> |

This report lacks a Report\_Thence field, so leave it blank in the spreadsheet

| Section 3: Point of Diversion and Legal Land Description                                                                                                                                                                                                                                                                                                                                          |                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Provide the location of the POD using one of the following methods (check one box and enter data if applicable)                                                                                                                                                                                                                                                                                   |                                               |
| <input checked="" type="checkbox"/> Latitude/Longitude Measurements: <div style="display: flex; justify-content: space-between;"> <div>             Latitude: <b>39.119400</b><br/>             Report_Latitude           </div> <div>             Longitude: <b>-123.119660</b><br/>             Report_Longitude           </div> </div>                                                        |                                               |
| <input type="checkbox"/> California Coordinate System (NAD1983) <div style="display: flex; justify-content: space-between;"> <div>             North: _____<br/>             Report_Northing           </div> <div>             East: _____<br/>             Report_Easting           </div> <div>             Zone: _____<br/>             Report_Zone           </div> </div>                   |                                               |
| <input type="checkbox"/> USGS Topographic Map with Point of Diversion labeled on map (if checked, map must identify point of diversion) <div style="text-align: right;">Report_Map</div>                                                                                                                                                                                                          |                                               |
| Assessor's Parcel Number (APN) where Point of Diversion is located (if APN has been assigned)<br><b>189-070-24-00</b><br>Report_APN                                                                                                                                                                                                                                                               | County<br><b>Mendocino</b><br>Report_Meridian |
| Provide Public Land Description to nearest 40 acres (if assigned) <div style="display: flex; justify-content: space-between;"> <div>             SW <math>\frac{1}{4}</math> of the NE <math>\frac{1}{4}</math> of Section <b>31</b> </div> <div>             Township <b>15N</b> </div> <div>             Range <b>11W</b> </div> <div>             B&amp;M Mount Diablo           </div> </div> |                                               |
| Report_QQ_Section                                                                                                                                                                                                                                                                                                                                                                                 | Report_QSection                               |
| Report_Section                                                                                                                                                                                                                                                                                                                                                                                    | Report_Township                               |
| Report_Range                                                                                                                                                                                                                                                                                                                                                                                      |                                               |

| Section 4: Map Description                                                                                                                                                                                                                                                                                           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| (Check boxes indicating each map to be provided)                                                                                                                                                                                                                                                                     |  |
| Identify the location of the place of use on a specific United States Geological Survey (USGS) Topographic Map, or County Assessor's parcel map or any other maps with identifiable landmarks. If assigned, provide the public land description to the nearest 40-acre subdivision and the assessor's parcel number. |  |
| <input type="checkbox"/> USGS Topographic Map <input type="checkbox"/> County Assessor's Parcel Map <input checked="" type="checkbox"/> Map with identifiable landmarks                                                                                                                                              |  |
| Provide a general description of the area in which the water was used (e.g. Domestic water supply for house, and irrigated crops, campground, etc.)                                                                                                                                                                  |  |
| <b>Domestic water supply</b>                                                                                                                                                                                                                                                                                         |  |
| Assessor's Parcel Number(s), where the water was used (if APNs have been assigned)<br><b>189-070-24-00</b>                                                                                                                                                                                                           |  |

| Section 5: Purpose of Use Description                         |                                                                                               |                                                                            |                                                  |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------|
| (Select all that apply)                                       |                                                                                               |                                                                            |                                                  |
| <input type="checkbox"/> Irrigation<br>Number of acres: _____ | <input checked="" type="checkbox"/> Domestic<br>Maximum number of persons served:<br><b>2</b> | <input type="checkbox"/> Stock watering<br>Number and type of stock: _____ | <input type="checkbox"/> Other<br>Explain: _____ |

S027304

Figure 33. Locating the Report Fields on an eWRIMS document (Part 2)

- 2) Note that not all eWRIMS PDFs have the same design—the water rights form has evolved over the years and different types of rights have different fields. Additionally, even after you apply optical character recognition (OCR) to these PDFs, you still may be

unable to apply Ctrl+ F to search for terms because of poor scan quality, especially for older water rights forms. Furthermore, you also do not need to fill out every single field that you added to your manual review spreadsheet with the *M script* (hosted in the *DWRAT\_DataScraping/Demand/Scripts* folder). If the report already explicitly states the latitude and longitude of the PODs, then you do NOT need to note all the other *supplemental* geolocational information. The purpose of this supplemental information is to derive the POD latitude and longitude if absent from the report.

### 3) Filling out the Fields in the Manual Review Spreadsheet

- a. *Report\_Latitude* and *Report\_Longitude*: If the eWRIMS report has the latitude and longitude, then fill out these two fields. If the coordinates are not provided in decimal degrees, then convert them to decimal degrees before entering them into the spreadsheet. The purpose of these fields is to gauge the accuracy of the eWRIMS GIS coordinates (*eWRIMS\_Latitude* and *eWRIMS\_Longitude*)—if *eWRIMS\_Latitude* and *eWRIMS\_Longitude* are accurate, then set the *LATITUDE* and *LONGITUDE* fields in the manual review spreadsheet equal to *eWRIMS\_Latitude* and *eWRIMS\_Longitude*. If they are inaccurate, then instead populate the *LATITUDE* and *LONGITUDE* fields with the *Report\_Latitude* and *Report\_Longitude*.
- b. *Report\_Northing* and *Report\_Easting*: Northings and eastings are coordinates in the state plane coordinate system. The California version of the state plan coordinate system is known as the *California Coordinate System*. If the eWRIMS report lacks the *Report\_Latitude* and *Report\_Longitude* fields, then look for the *Report\_Northing* and *Report\_Easting*. Check that the diverter reported the northing and easting in units of feet and note the *Report\_Zone* as well. The *Report\_Zone* is always established with respect to a coordinate reference system (CRS), although this CRS may not be explicitly stated.

#### Explicit CRS example

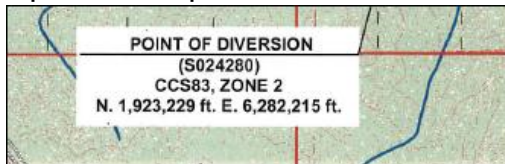


Figure 34. CCS83, ZONE 2 indicates that the CRS is NAD 1983—the CCS is short for California Coordinate System.

#### Report\_Zone with no CRS

Occasionally reports do not indicate the CRS (Payman Alemi was unable to find an example after a cursory search but has seen it happen before). However, the CRS can usually be determined based on context. Northing and easting can be provided with either NAD 1927 or NAD 1983 CRS, but if you convert both sets of coordinates into WGS 84 and plot them in ArcGIS Pro against your watershed polygon shapefile, you can determine which CRS makes more sense. If no *Report\_Zone* is listed in the report, then you will need to rely on other geolocational parameters to back-calculate the *Report\_Latitude* and *Report\_Longitude*. For detailed instructions on this back-calculation process, refer to Appendix [3.3](#).

- c. *Report\_Source\_Creek*, *Report\_Tributary*, and *Report\_Thence* fields: The purpose of filling out these fields is to assess whether the POD flows into a stream that belongs to the watershed of interest. If none of these fields have “Russian River” for example, then it is likely that the POD flows into another watershed. As an extra check, enter the *Report\_Latitude* and *Report\_Longitude* into the USGS StreamStats tool.
- d. *Report\_Meridian*, *Report\_APN*, *Report\_Township*, *Report\_Range*, *Report\_Section*, *Report\_QSection*, *Report\_QQSection*, and *Report\_POD\_Information* are only necessary to populate when the report lacks any type of POD coordinates, or you suspect that the coordinates are incorrect. Additionally, not every report may have all these fields, but even with a handful of them, you can roughly pinpoint the POD coordinates by using ArcGIS Pro. Appendix 3.4 provides a detailed example of this process by using the POD for water right A028127.

In a nutshell, the steps above safeguard against the possibility that the coordinates in eWRIMS were radically misplotted—our ArcGIS Pro analysis has already determined that the eWRIMS GIS coordinates are located within the watershed. However, if the report coordinates differ substantially from the eWRIMS GIS coordinates, we must check that they are still located within the watershed. If they are NOT located within the watershed, then type “No” in the *Keep\_POD* field in the manual review spreadsheet because we want to exclude these PODs from the final demand dataset.

- e. *StreamStats* field—Fill out this field with “In watershed” or “Outside watershed” depending on the results of the USGS StreamStats Tool (see Appendix 3.2) The purpose of this field is to catch PODs that are geospatially within the watershed but that flow into another watershed—this is rare but possible. All “outside watershed” records should have their corresponding *Keep\_POD* field set to “No” because they should also be excluded from the final demand dataset.
  - f. *Keep\_POD*: Indicate as “Yes” or “No”—you either keep the POD or exclude it from the final POD list for the watershed.
  - g. *POD\_Justification*: Justify your decision to keep or exclude the POD and be as detailed as possible (cite coordinate system discrepancies and other issues you came across for example)
  - h. *Staff*: Type your name.
  - i. *Review\_Date*: Type the date you conducted the manual review for the POD.
- 4) Keep in mind that the manual review spreadsheet that you created from ArcGIS Pro contains only a subset of the PODs in the watershed—this is the subset that might possibly have geolocational errors. Consequently, you need to remove the errant PODs you identified in the previous steps from the full POD list in the *XX\_pods\_merged\_YYYY-MM-DD* csv.
- a. Using Power Query, import the *XX\_pods\_merged\_YYYY-MM-DD* csv into your *manual review spreadsheet*.
  - b. Remove all PODs that were identified for removal during the manual review process and save this filtered list as a separate sheet called *Final\_List*.

- 5) Type the full name of the *[ID]\_GIS\_Preprocessing.xlsx* spreadsheet and the *Final\_List* in *Watershed\_Demand\_Datasets\_Path.xlsx*:
  - a. POD\_APPLICATION\_NUMBER\_SPREADSHEET\_PATH: this is the file path of your manual review spreadsheet.
  - b. POD\_APPLICATION\_NUMBER\_WORKSHEET\_NAME: set to “Final\_List”
  - c. IS\_SHAREPOINT\_PATH\_POD\_APPLICATION\_NUMBER\_SPREADSHEET: set to “FALSE”.
- 6) You can now jump to **Section 7. Preliminary Demand Dataset** of the main [Demand Dataset Procedure](#) document.

### 3 Appendix

#### 3.1 Using Power Query and M Scripts

1. Navigate to **Data → Get Data → Launch Power Query Editor**

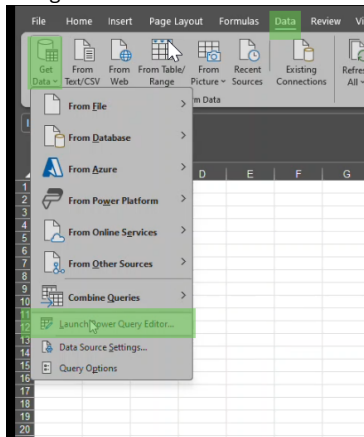


Figure 35. Launch Power Query Editor

2. Navigate to the **Home Tab → New Source → File → Text/CSV** and load the manual review spreadsheet that you generated in ArcGIS Pro [*XX\_pods\_manual\_review\_YYYY-MM\_DD.csv*] by clicking **Import** and then **OK** on the resulting dialog boxes.

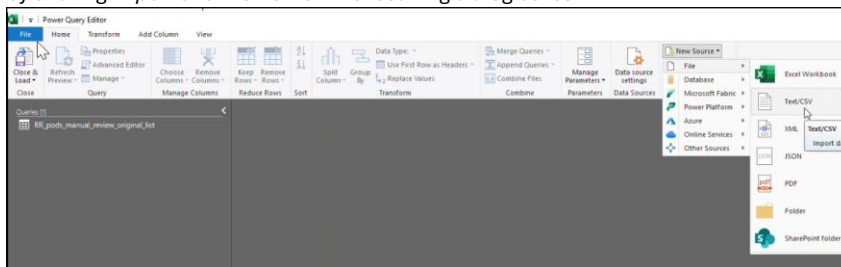


Figure 36. Importing CSVs into Power Query

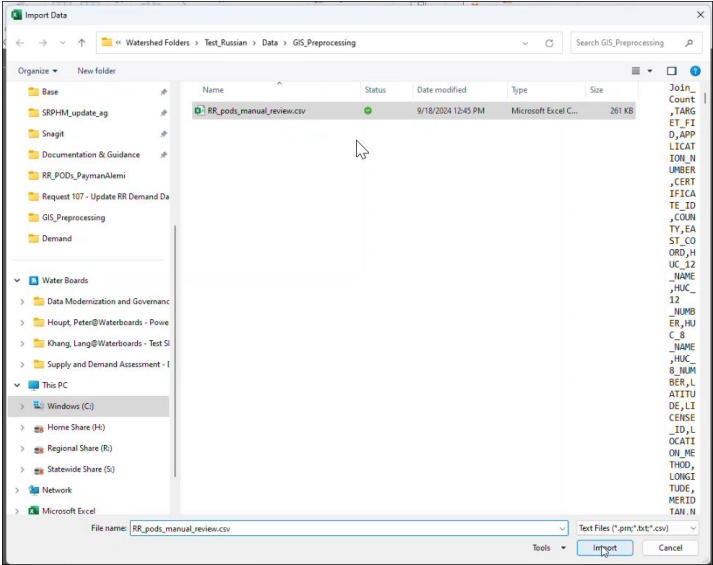


Figure 37. Import Data Window

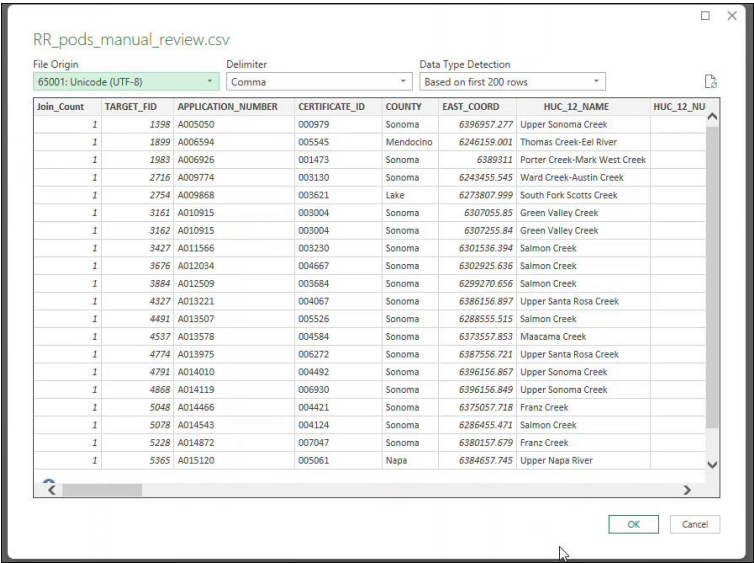


Figure 38. Loading RR\_pods\_manual\_review.csv into Power Query Editor

3. Rename the CSV ("query") that you just imported to *XX\_pods\_manual\_review\_original* ("RR\_pods\_manual\_review\_original" in this example) by right-clicking on it.

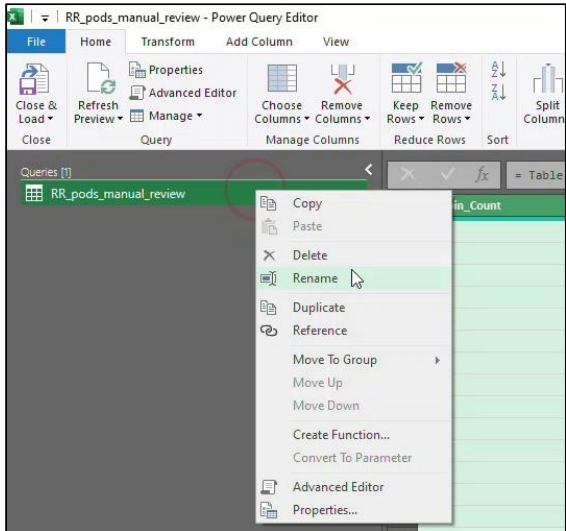


Figure 39. Renaming the query

4. Right-click on the query again and select *Advanced Editor*.
5. Delete everything from the resulting *Advanced Editor* window:

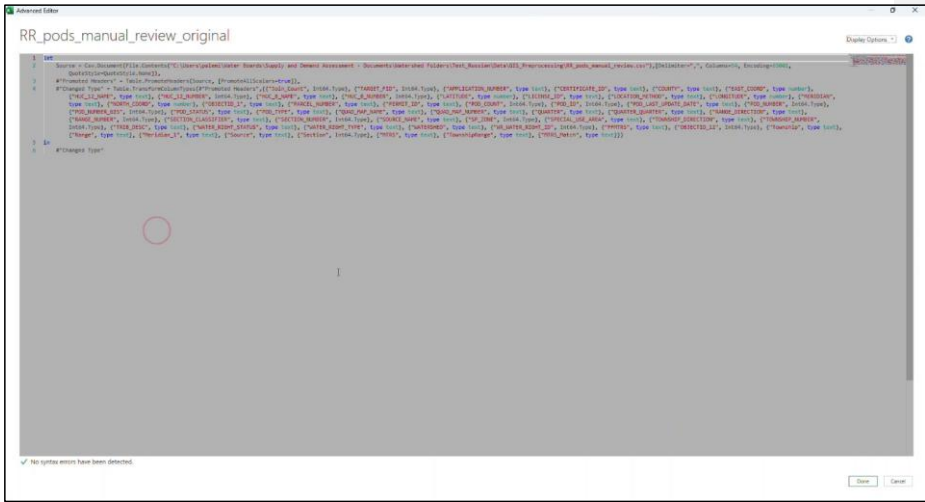


Figure 40. Advanced Editor in Power Query Editor

6. Load the M script of interest (“RR\_GIS\_Manual\_Review\_M\_Script.txt” in this example) in Notepad++.
7. Copy and paste all the content from the M Script into the *Advanced Editor* window.
8. Adjust the portion of the M Script that defines the local user folder to point to yours:  
**let**

## 31 of 38

Source = Csv.Document(File.Contents("C:\Users\palemi\Water Boards\Supply and Demand Assessment - Documents\Watershed Folders\Test\_Russian\Data\GIS\_Preprocessing\RR\_pods\_manual\_review.csv"))

9. Confirm there are no syntax errors and click *Done*.
10. Load the query as a sheet with the same name into Excel by navigating to *Home* → *Close and Load* in the ribbon.

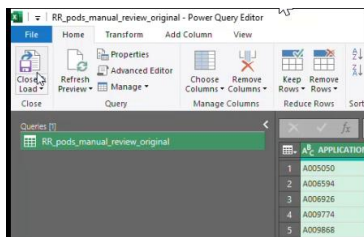


Figure 41. Close and Load the query

### 3.2 Using the USGS StreamStats Tool

- 1) Go to <https://streamstats.usgs.gov/ss/> and enter the latitude and longitude of the POD of interest into the search box:

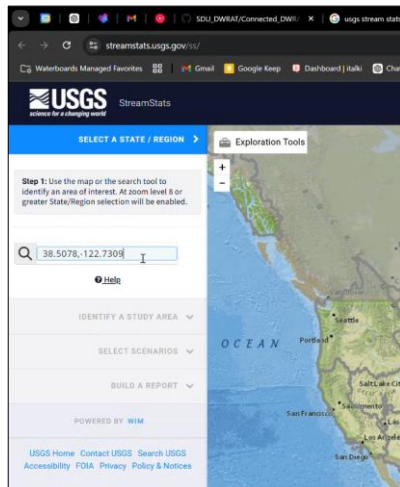


Figure 42. Entering Coordinates in StreamStats

- 2) Then select the automatic coordinates that appear in the dropdown menu:

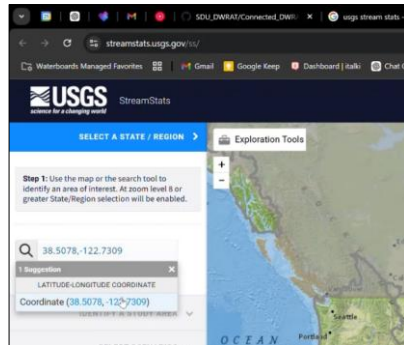


Figure 43. Coordinate Dropdown option

- 3) Next click *Exploration Tools* and scroll down to *Flow (Raindrop) Path*:

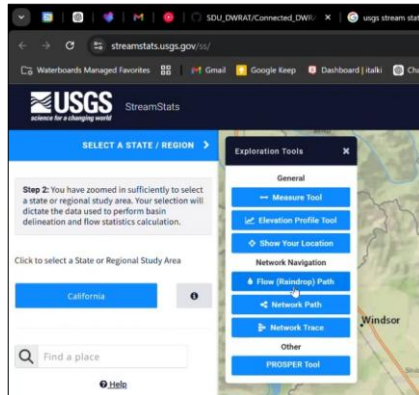


Figure 44. Flow (Raindrop) Path

- 4) After that, click *Select Point on Map*, pan to the map and then select the red point.

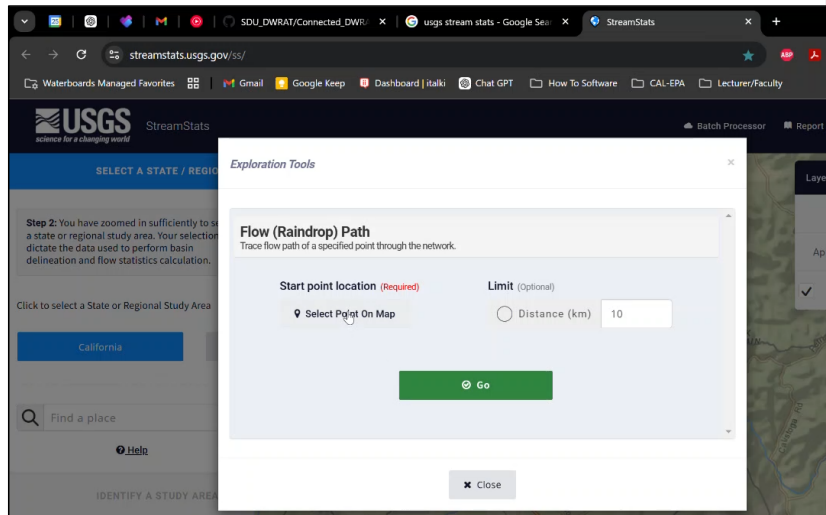


Figure 45. Flow (Raindrop) Path



Figure 46. Selecting the point on the map

5) The Flow (Raindrop) Path pop-up menu will re-appear and now click Go.

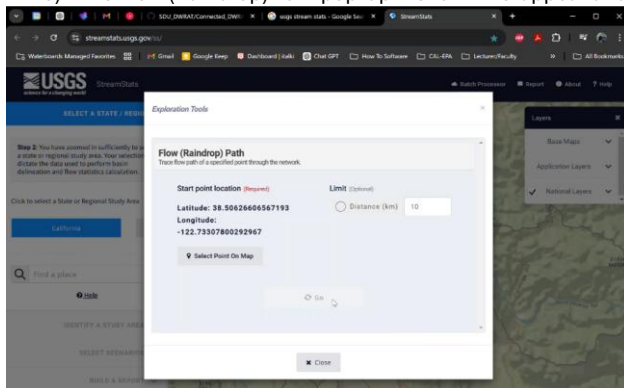


Figure 47. Flow (Raindrop) Path—Click Go

6) Zoom to the end of the red flow path, read the map carefully and confirm that the final watercourse name matches the name of the watershed. In this example, the

final watercourse name is the *Russian River*, which confirms that the POD flows into the *Russian River* watershed:



Figure 48. Watershed Outlet in StreamStats

### 3.3 Using EarthPoint to convert Northings/Easting into Latitude/Longitude

- 1) Browse to <https://www.earthpoint.us/StatePlane.aarcgisdspx> and enter these parameters and click *Calc.*  
*XY Unit Measure:* Us Survey Feet  
*Zone Number:* 0402 or select 0402 – California Zone 2 from the dropdown list.  
*Easting/X:* 6282215  
*Northing/Y:* 1923229
- 2) Scroll down the page to find the latitude and longitude in decimal degrees:

|                                                                        |                                        |
|------------------------------------------------------------------------|----------------------------------------|
| Calculated Values - based on Degrees Lat Long to seven decimal places. |                                        |
| Position Type                                                          | State Plane - California Zone 2        |
| Degrees Lat Long                                                       | 38.4390900°, -122.9756534°             |
| Degrees Minutes                                                        | 38°26.34540', -122°58.53921'           |
| Degrees Minutes Seconds                                                | 38°26'20.7239", -122°58'32.3524"       |
| State Plane X Y (Meters)                                               | 0402 1915822.962mE 586201.372mN        |
| X Y (US Survey Feet)                                                   | 0402 6282215.000UsFtE 1923229.000UsFtN |
| X Y (International Feet)                                               | 0402 6282227.564ftE 1923232.846ftN     |
| UTM                                                                    | 10S 502124mE 4254535mN                 |
| UTM centimeter                                                         | 10S 502124.75mE 4254535.07mN           |
| MGRS                                                                   | 10SEH0212454535                        |
| Grid North                                                             | 0.0°                                   |
| GARS                                                                   | 115LS11                                |
| Maidenhead                                                             | CM88MK25WJ11                           |
| GEOREF                                                                 | DJNJ01462634                           |
| Plus Code                                                              | 84CVC2QF+JP                            |
| Plus Code Extended                                                     | 84CVC2QF+JPQCVVW                       |
| what3words                                                             | relief.bargained.chatting              |

Figure 49. Northing/Easting Conversion to Latitude and Longitude with Earthpoint

- 3) In ArcGIS Pro, confirm if your coordinates make sense and match the existing eWRIMS coordinates—navigate to the *Map* tab and click on the *Go to XY* entry tool.

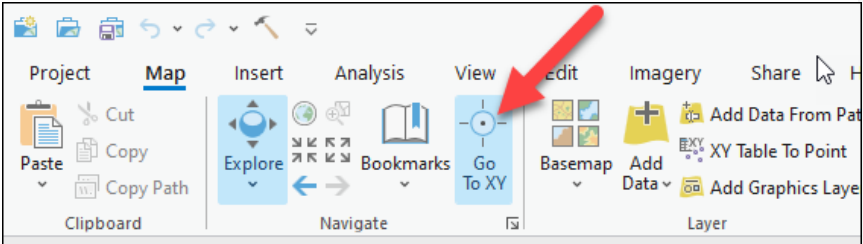


Figure 50. Go to XY Tool

- 4) Add an XY label to the map and flash the point. Use the pan and zoom features to bring the point into view. Select the [WS]\_pods\_manual\_review shapefile and look for the nearest PODs. In this case, as [Figure X] demonstrates, the report coordinates match identically with the eWRIMS coordinates for the water right S024280.

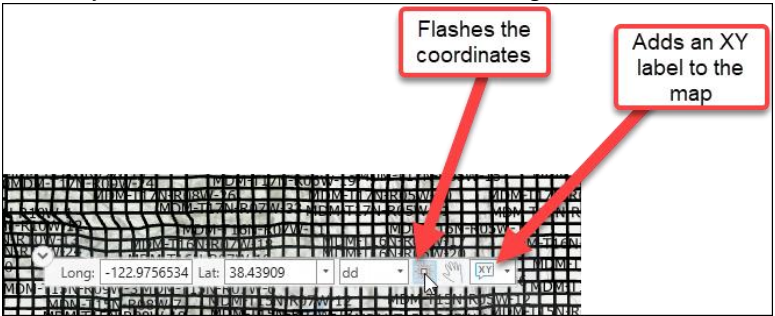


Figure 51. Features of XY Tool

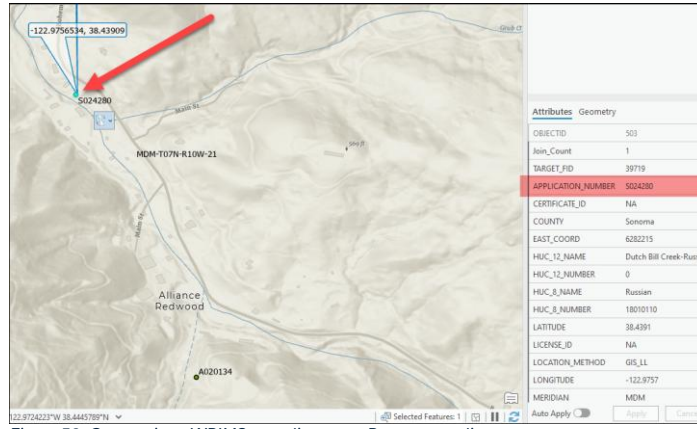


Figure 52. Comparing eWRIMS coordinates to Report coordinates

- 5) Type the latitude and longitude that you derived from the EarthPoint website into the *Report\_Latitude* and *Report\_Longitude* fields but indicate with comments (right-click on each cell to create a comment) that these values were not listed in the report but were derived from the Northing and Easting.

### 3.4 Finding POD Coordinates in the absence of latitude/longitude and northing/easting

- 1) Taking water right A028127 as an example, we open the report and look for the POD location description.

THE POINT OF DIVERSION OF SUCH WATER IS LOCATED:

North 1,000 feet and West 600 feet from SE corner of Section 17, T17N, R13W, MDBM, being within SE¼ of SE¼ of said Section 17.

Figure 53. A028127 POD location description

In this example, no Report\_APN is listed but we have sufficient information to pinpoint the POD in eWRIMS because we have values for these fields:

- Report\_Meridian: MDBM (“Mount Diablo Base and Meridian”)
- Report\_Township: T17N (or “17N”)
- Report\_Range: R13W (or “13W”)
- Report\_Section: 17
- Report\_QSection: SE (“Southeast”)
- Report\_QQSection: SE
- Report\_POD\_Information: North 1000 ft and West 600 ft from SE corner of Section 17, 17N, 13W MDBM—you can use these measurements to pinpoint the exact location in ArcGIS Pro, but you don’t have too—eyeballing the QQSection is sufficient.

- 2) Make sure that you have loaded these shapefiles into ArcGIS Pro:
  - *RR\_QQ\_Sections* (this is derived from a larger statewide QQ\_Section shapefiles but I whittled it down to the Russian River for this example)

### 37 of 38

- *PLSS\_Sections\_Fill\_BLM\_2024\_NAD\_1983* (this is the PLSS layer for the entire state of California)
- *RR\_pods\_manual\_review*

3) You can expedite the process by assuming that the coordinates plotted in eWRIMS are accurate (even though the report itself lacks coordinates); the vast majority of the time, eWRIMS coordinates are correct. This allows us to open the attribute table of the *RR\_pods\_manual\_review* shapefile, select the record for A028127 and zoom to the selected record:

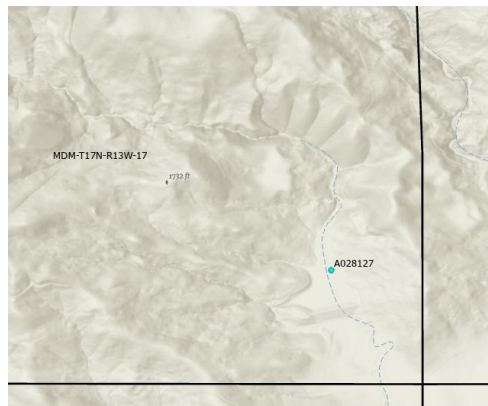


Figure 54. A028127 POD superimposed on statewide PLSS layer

Figure 54 confirms that the POD is indeed located in Section 17 of Township 17N, Range 13W. If we zoom in, we can visually confirm, as shown in Figure 55, that this POD is located within the SE quarter of Section 17.

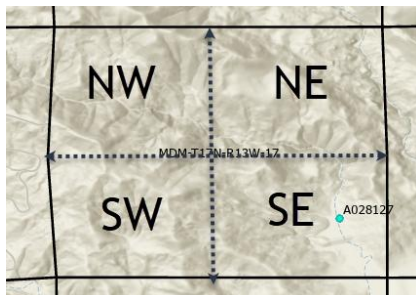


Figure 55. Dividing Section 17 into Quarter-Sections

After we load the *QQ\_Sections* shapefile, and zoom in further, as shown in Figure X, we visually confirm that the POD is located within SE quarter of the SE quarter-section.

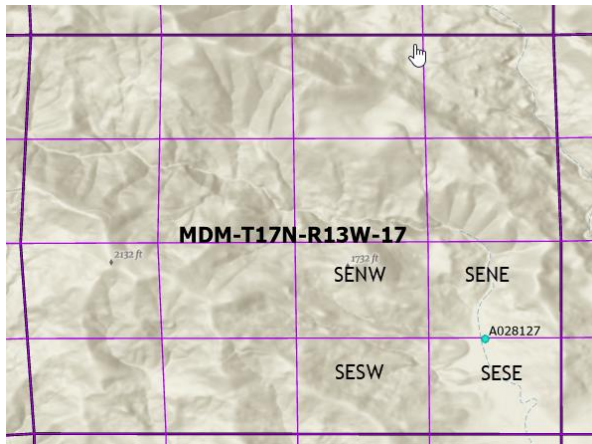


Figure 56. A028127 superimposed on quarter-quarter sections

For our purposes, this is accurate enough and it validates our initial assumption that the eWRIMS coordinates are accurate. Consequently, we can set the final LATITUDE and LONGITUDE fields in the manual review spreadsheet equal to the eWRIMS\_Latitude and eWRIMS\_Longitude fields.